Aegis-v2: Detection-Gated, Layer-Selective Self-Repair of Onboard Earth-Observation Classifiers under Orbit-Calibrated Radiation Upsets

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***Abstract*—Deep neural networks now run on commercial off-the-shelf (COTS) processors aboard Earth-observation satellites, where single-event upsets (SEUs) from trapped protons silently corrupt the stored weights. In IEEE-754 single precision, one flip of the most significant exponent bit turns a weight of order one into a value near 10^{38} and collapses a classifier to chance. Existing remedies sit at two extremes: blind periodic scrubbing rewrites the whole model on a timer, while in-situ software mitigation (activation clipping, batch-norm re-centering) leaves corrupted weights in memory, where they accumulate. We present Aegis-v2, a detection-gated, layer-selective repair loop. A label-free behavioral monitor (an activation-range envelope and a batch-norm drift test) proposes suspect blocks on each incoming image batch; a CRC-32 comparison against golden checksums confirms which blocks actually changed, including the block just upstream of the earliest alarm; only confirmed blocks are restored, bit-exactly, from a compressed golden copy in radiation-tolerant storage. An optional periodic CRC audit catches faults too small to change behavior. We evaluate Aegis-v2 in a paired, seeded mission simulator that flies a real Sentinel-2A track (SGP4), defines the South Atlantic Anomaly physically from IGRF field strength, and calibrates upset rates to eight years of Alsat-1 flight data. Over three simulated days and three radiation seeds on EuroSAT, Aegis-v2 keeps a ResNet-18 at 87.4% mean accuracy (clean 89.3%) and within five points of clean for 89% of the mission, against 37.0% and 26% for 12-hour scrubbing, 38.5% for range clipping and 14.0% unprotected, winning every paired seed; matching its accuracy by scrubbing would take $4.7\times$ the repair traffic. A held-out-trained MobileNetV2 keeps clean accuracy at 3.4 MB/day. An exact ablation on identical fault streams shows that the upstream-block rule prevents a 35-hour outage. In 300 controlled single upsets, only exponent-MSB flips caused damage, and every damaging flip was detected and localized with no false restore; damaged classifiers collapse toward a single, fault-dependent land-cover class. We also report a negative result: always-on monitoring makes Aegis-v2 cost 5–$22\times$ more energy than slow scrubbing under our energy model, so its advantage is bandwidth and availability, not energy. We document the evaluation pitfalls that invalidated our own early results and release the simulator, campaigns and analysis scripts.**

***Index Terms*—Single-event upset, soft errors, fault tolerance, onboard machine learning, Earth observation, deep neural networks, self-repair, South Atlantic Anomaly, memory scrubbing, batch normalization.**

I. Introduction

ONBOARD inference has moved from demonstration to routine. Φ-Sat-1 flew a convolutional cloud screener on a COTS vision accelerator in 2020 and discarded cloudy frames before downlink [1]; OPS-SAT trained and replaced models in orbit [2]; current small-satellite payloads carry Jetson-class GPUs and neural accelerators [3], [4]; and system papers now treat AI services in orbit as infrastructure [5]. The appeal is plain: the downlink is the bottleneck of Earth observation, so deciding onboard what is worth sending multiplies the useful data a satellite returns. The cost is reliability. COTS processors and memories are not radiation hardened, and in low Earth orbit (LEO) energetic protons trapped in the inner Van Allen belt, concentrated over the South Atlantic Anomaly (SAA) [6], deposit enough charge to flip stored bits [7], [8].

A neural network is an unusually fragile thing to store in such memory. Its knowledge lives in millions of floating-point weights, and the IEEE-754 binary32 format [9] packs an 8-bit exponent beside the sign. Trained weights are small, so their exponent's most significant bit is almost always zero; flipping it multiplies the weight by $2^{128} \approx 3.4 \times 10^{38}$. One such upset in a convolution kernel saturates every activation downstream and drives the classifier to chance, as targeted bit-flip attacks showed on purpose [10] and statistical injection studies confirm for vision transformers [11]. Upsets also accumulate: unlike a transient glitch in logic, a flipped bit in SRAM or DRAM stays flipped until something rewrites it.

Spacecraft engineering answers with hardware: error-detecting and correcting codes (EDAC), triple modular redundancy and memory scrubbing, the periodic rewriting of memory from a trusted copy [8], [12], [13]. Scrubbing is simple and robust but blind: it pays to rewrite the whole model whether or not anything is wrong, and between scrubs the model runs corrupted. The machine-learning community answers with software. Range restriction (Ranger [14], FT-ClipAct [15]) clamps activations to profiled bounds; BLINK [16] detects batch-norm drift and re-centers activations; RADAR [17] and MILR [18] detect or algebraically recover corrupted weights; fault-aware training hardens the weights themselves [19]. These methods mitigate symptoms in place but, with the exception of algebraic recovery, leave the corrupted weights in memory, where further upsets compound them. A third option is to wait for a ground station and re-upload the model, which costs scarce uplink time and leaves the model broken until the next contact. Fig. 1 contrasts the four behaviors on one fault sequence.

This paper asks whether a satellite can do better by combin-

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Code, campaign results, the analysis scripts that generate every table and figure, and an interactive mission dossier are available in the project repository (`Rg9906/aegis-v2`).

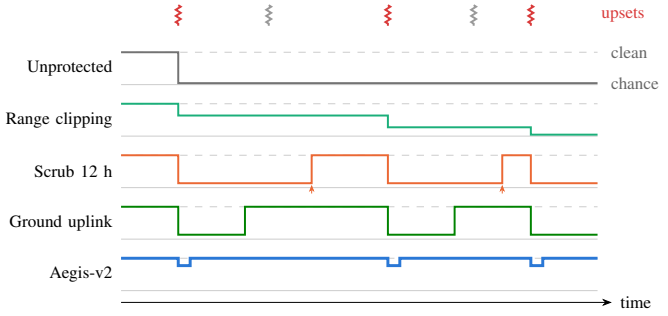


Fig. 1. Schematic behavior of the strategies on one sequence of upsets (red: damaging; grey: harmless). Clipping softens each hit but cannot undo it; scrubbing recovers only on its timer; a ground reload only at contacts; Aegis-v2 detects and restores the damaged block at the next monitored batch and ignores harmless hits.

ing the two worlds: use cheap, label-free signals from the network's own activity to notice damage, use exact checksums to prove which stored blocks changed, and then restore only those blocks from an onboard golden copy. We call this *detection-gated, layer-selective restore* and build it as Aegis-v2. The idea is simple; demonstrating it honestly is not. Evaluations of DNN resilience are easy to get wrong in ways that flatter the method: unrealistic fault models (zeroed weights instead of exponent flips) [20], synthetic inputs, monitors whose thresholds fire on every clean batch, energy budgets that omit the monitor itself. Our own first prototype reported a 98.9% energy saving that did not survive correction (Section VIII). We therefore treat the evaluation methodology as a first-class contribution.

Contributions.

1) **A detection-gated, layer-selective repair loop** (Aegis-v2) that combines a two-tier label-free behavioral monitor with CRC-32 confirmation, an *upstream-block rule* that localizes batch-norm parameter faults whose symptoms appear one block downstream, and an optional periodic CRC audit (Section V).
2) **A paired, orbit-calibrated mission simulator**: faults placed uniformly over stored bits at rates integrated minute by minute along a real Sentinel-2A track, with the SAA defined from IGRF field strength and the orbit-mean rate calibrated to Alsat-1 flight data; every strategy replays the identical fault stream (Section IV).
3) **A head-to-head evaluation** on two architectures against no protection, range clipping, BN re-centering, periodic scrubbing at five intervals and a ground-uplink reload, reporting mean accuracy, availability, accuracy-loss area, repair traffic and modeled energy with 95% intervals and paired differences (Section VII).
4) **An exact ablation and a single-fault detection study**: the upstream rule is isolated on identical fault streams, and 300 controlled single upsets measure what each tier detects, by bit field, and what a damaged classifier predicts (Sections VII-F–VII-H).
5) **An honest cost accounting with a negative result**: with monitor compute charged, Aegis-v2 is not an energy saver (Section VII-J).

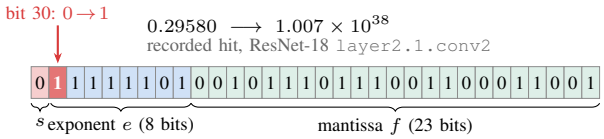


Fig. 2. IEEE-754 binary32 layout of a weight hit during our campaign (ResNet-18, seed 1, hour 2). Flipping the most significant exponent bit multiplies the value by 2^{128}. Mantissa flips, 72.5% of all recorded hits, change the value by at most a factor of two.

6) **A catalogue of evaluation pitfalls**, each of which silently invalidated a result during this work, with the invariant that prevents it (Section VIII).

II. Background

A. Radiation in Low Earth Orbit

In LEO the dominant source of SEUs in memory is the flux of trapped protons in the inner radiation belt, supplemented by galactic cosmic rays and solar particle events [8]. Because Earth's dipole is offset and tilted, the inner belt reaches its lowest altitude over the South Atlantic, where the field strength at a given altitude is weakest; this is the SAA [6]. A sun-synchronous satellite crosses it on several consecutive orbits each day, and most memory upsets on such missions are recorded inside it. Flight missions handle this with EDAC and background scrubbing of program memory; the Flying Laptop mission documents in-flight SEU statistics and scrubbing operations [13].

The Alsat-1 record is unusually useful for calibration: between November 2002 and August 2010 its commercial SRAM logged 265,649 upsets in 2,622 days, i.e. 4.04×10^{-7} upsets/bit/day, of which 98.55% were single-bit upsets (SBUs) and the remainder multiple-bit upsets (MBUs); the yearly mean rose from 41 to 119 upsets per day over the solar cycle [12].

B. Why Floating-Point Weights Are Fragile

A binary32 value is $v = (-1)^s\, 2^{e-127}\,(1+f)$ with sign s, biased exponent $e \in [1, 254]$ for normal numbers, and fraction $f \in [0, 1)$ (Fig. 2). Table I summarizes what one flip does. A flip in mantissa bit k changes $1+f$ by $\pm 2^{k-23}$, a factor between $2/3$ and $3/2$ at worst, so most mantissa flips are invisible to a classifier. A flip in exponent bit $j \in [23, 30]$ multiplies v by $2^{\pm 2^{j-23}}$. Trained weights satisfy $|v| < 2$, so $e \le 127$ and bit 30 is zero; a flip sets it and multiplies the value by exactly 2^{128}, producing either a huge finite value or $\pm\infty$. A sign flip on a batch-norm running variance makes $\sqrt{\sigma^2 + \epsilon}$ undefined and yields NaN. Under uniform placement about one hit in 32 lands on bit 30; we recorded 4.89 such hits per day on ResNet-18 against 4.6 expected.

C. Onboard Earth-Observation Classification

We use land-use/land-cover classification of Sentinel-2 [21] imagery as the workload, with the EuroSAT benchmark [22]: 27,000 RGB chips of 64×64 pixels in ten classes (Annual Crop, Forest, Herbaceous Vegetation, Highway, Industrial,

TABLE I
EFFECT OF A SINGLE FLIP BY BIT FIELD IN A TRAINED BINARY32 WEIGHT ($|v| < 2$).

Field (bits)	Share	Multiplier on $\lvert v\rvert$	Typical consequence
Sign (31)	$1/32$	1 (sign change)	NaN if on a BN variance
Exponent MSB (30)	$1/32$	2^{128}	activation explosion
Exponent (23–29)	$7/32$	$2^{\pm 1} \ldots 2^{\pm 64}$	small to catastrophic
Mantissa (0–22)	$23/32$	$[2/3,\ 3/2]$	usually negligible

Pasture, Permanent Crop, Residential, River, Sea/Lake). The models are a ResNet-18 [23] and a MobileNetV2 [24], both with batch normalization [25], the two families most often proposed for onboard vision.

III. RELATED WORK

Hardware mitigation. EDAC/ECC, selective or full TMR, and scrubbing remain the default for flight software memory [8], [12], [13]. ECC adapted to DNN weights corrects single-bit errors and masks uncorrectable weights, at 47.5% area overhead [26]; Safe-NEureka adds dual-modular redundancy with hardware recovery to a satellite DNN accelerator, protecting memory with ECC and the controller with TMR [27]; ZEDAC protects model parameters in memory and storage with row/column extended-Hamming codes held in a rad-tolerant companion, restoring corrupted parameters without retraining when errors stay within its correctable regime [28]. Beam experiments caution that up to 61% of DNN errors on GPUs were critical, that ECC did little to reduce them, and that single-bit injection underestimated the measured misdetection rate of YOLOv3 by $8.66\times$ [20]. Aegis-v2 is complementary: it assumes a protected golden copy and repairs the unprotected working copy.

Algorithm-based fault tolerance and checksums. ABFT [29] and its convolutional descendants detect computation errors via checksums; RADAR [17] compares group checksums of weights against golden signatures and zeroes flagged groups; weight reconstruction [30] defends against targeted bit-flip attacks, which, like radiation, are most damaging in the exponent MSB [10]. We use the simplest member of this family, CRC-32 [31] per block, but only as a confirmer gated by behavior.

Range restriction. Ranger [14], FT-ClipAct [15] and ProAct [32] clamp activations to bounds profiled on clean data; Ranger reports 3–50$\times$ resilience gains for transient faults at negligible cost. They need no storage, but they do not repair weights; under accumulating upsets the clipped network degrades, which our ResNet-18 results confirm. RedNet combines such tolerance with application awareness and early exit, and was validated on the Chaohu-1 SAR satellite payload [4].

Batch-norm based detection. BLINK [16] uses batch-norm statistics as integrity checkpoints, reporting over 99% detection precision and recovering ResNet-50/CIFAR-10 from 10% to 85.9% under 0.5% random bit flips with under 2% latency overhead. Our Tier 2 is BLINK-style, and our BLINK baseline implements its re-centering mitigation. The difference is what happens next: Aegis-v2 restores weights instead of compensating for them.

TABLE II
CAPABILITY COMPARISON. (A) LABEL-FREE BEHAVIORAL MONITOR; (B) RESTORES CORRUPTED WEIGHTS; (C) NO LABELS OR GROUND CONTACT NEEDED FOR RECOVERY; (D) EVALUATED UNDER ORBIT-CALIBRATED ACCUMULATING FAULTS.

Approach	(a)	(b)	(c)	(d)
Periodic scrubbing [8]	–	blind	✓	–
ECC / ZEDAC [26], [28]	–	on read	✓	–
Ranger / FT-ClipAct [14], [15]	bounds	–	✓	–
BLINK [16]	BN	–	✓	–
RADAR [17]	–	zeroes group	✓	–
MILR [18]	–	algebraic	✓	–
CoG correction [33]	–	substitutes	✓	–
RedNet [4]	–	–	✓	flight
Ground re-upload	–	late	–	–
Aegis-v2 (ours)	**bounds+BN**	**CRC-gated**	✓	✓

Weight correction, hardening and analysis. MILR [18] recovers layers algebraically from stored input/output relations; CoG-guided correction [33] substitutes profiled values for detected faulty weights, reporting up to $49.6\times$ fault-tolerance improvement on ResNet-18; fault-aware training improves tolerance to multiple SEUs by up to $3\times$ [19]. Layer- and bit-wise sensitivity analyses exist for semantic segmentation [34] and for vision transformers on EuroSAT, where only about 3% of FP32 flips cause failure but most failures are severe and concentrate in normalization layers and exponent bits [11]. Plasticity methods such as ReDo [35] and continual backpropagation [36] reinitialize dormant units during training; they cannot restore a specific learned function after hardware faults.

Space ML robustness and tooling. Lange et al. surveyed the robustness of onboard ML to radiation, found that not all faults are as damaging as prior work claimed, and reported that existing injection tools were hard to apply to satellite models [37]. Fault injectors such as PyTorchFI [38] and Ares [39] established application-level injection as a valid proxy.

Table II summarizes the gap. Every component of Aegis-v2 exists in isolation; to our knowledge the combination, evaluated as a repair *policy* under accumulating, orbit-modulated faults with paired baselines and traffic and energy accounting, has not been reported.

IV. FAULT MODEL AND MISSION SIMULATOR

A. What Is Corrupted

Let the model's stored state be $\theta \in \mathbb{F}_{32}^{N}$, all parameters and batch-norm buffers, occupying $B = 32N$ bits: 358.1 Mbit for ResNet-18 (44.77 MB) and 72.7 Mbit for MobileNetV2 (9.08 MB). We model SEUs in the working copy of θ held in volatile memory. Faults persist until a repair overwrites them. We do not inject into activations, control logic or the golden copy, which we assume resides in radiation-tolerant non-volatile memory (MRAM or rad-hard SPI NOR); transient activation faults are studied separately (Section VII-I).

B. Upset Arrival Along a Real Orbit

The satellite follows Sentinel-2A (NORAD 40697, $\approx$ 786 km sun-synchronous) propagated with SGP4 [40] from a

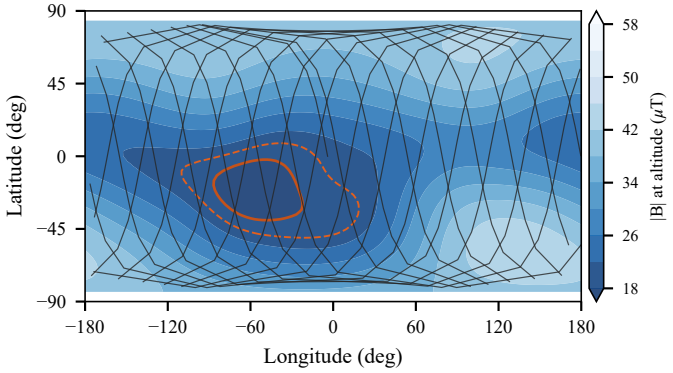


Fig. 3. One day of the Sentinel-2A ground track (SGP4) over IGRF field magnitude at altitude. Solid and dashed contours mark the SAA core (18,000 nT) and edge (20,000 nT) between which the upset rate ramps.

TABLE III
FAULT STATISTICS OF THE FINAL CAMPAIGNS (THREE SEEDS, THREE DAYS EACH). SEVERITY USES THE RULE OF SECTION VII-E.

	ResNet-18	MobileNetV2
Stored bits (Mbit)	358.1	72.7
Predicted flips/day ($B\bar{\lambda}$)	144.7	29.4
Recorded flips/day	147.1±16.6	29.1±0.5
Exponent-MSB flips/day	4.89±2.08	1.11±0.96
MBU events (% of events)	2.2	1.6
Flips in steps with SAA exposure (%)	90	90
Mantissa / exponent / sign (%)	72.5 / 24.6 / 2.9	73.7 / 22.9 / 3.4
Destructive / moderate / negligible (%)	3.4 / 18.7 / 77.9	3.8 / 16.4 / 79.8
Total recorded hits (3 seeds)	1,324	262

CelesTrak two-line element set [41], sampled every minute. At each position $\mathbf{r}(t)$ we evaluate the geomagnetic field magnitude $|B(\mathbf{r})|$ from IGRF-13 [42] and map it to a per-bit upset rate by a linear ramp:

$$\lambda(t) = \lambda_{\mathrm{bg}} + (\lambda_{\mathrm{SAA}} - \lambda_{\mathrm{bg}})\,\mathrm{clip}\left(\frac{B_{\mathrm{e}} - |B(\mathbf{r}(t))|}{B_{\mathrm{e}} - B_{\mathrm{c}}}, 0, 1\right), \quad (1)$$

with edge $B_{\mathrm{e}} = 20{,}000\,\mathrm{nT}$ and core $B_{\mathrm{c}} = 18{,}000\,\mathrm{nT}$. The number of upsets in a step of length Δ is Poisson with mean $B\int_t^{t+\Delta}\lambda(\tau)\,d\tau$, integrated per minute. Fig. 3 shows one day of the track over the field and Fig. 4 the resulting rate: background most of the day, punctuated by roughly ten short SAA passes, several of which reach the core rate.

Calibration. We fix the ratio $\lambda_{\mathrm{SAA}}/\lambda_{\mathrm{bg}} = 75.3$ and choose $\lambda_{\mathrm{bg}} = 7.89 \times 10^{-8}$ and $\lambda_{\mathrm{SAA}} = 5.94 \times 10^{-6}$ upsets/bit/day so that the orbit average over the real track (79.9 SAA-core-equivalent minutes per day) reproduces the Alsat-1 mean of 4.04×10^{-7} [12]. About 82% of upsets then occur inside the SAA. Table III checks the calibration against the campaigns: the predicted $B\bar{\lambda}$ matches the recorded flip rate on both architectures, and the recorded field distribution matches uniform placement.

C. Placement and Multiplicity

Each upset event is an SBU with probability 0.9855, otherwise an MBU cluster of up to four adjacent bits. Events are placed uniformly over stored bits: a block is chosen with probability proportional to its bit count, then a tensor in proportion to its size, an element and a bit. All 32 bit positions are eligible; restricting injection to the exponent would inflate severity.

D. Paired Replay

A single seeded injector samples the fault stream for the unprotected replica, recording (block, tensor, index, bit) tuples. Every other strategy runs on its own deep copy of the model and *replays the identical records* (Fig. 5). Differences between strategies within a seed are therefore paired, and we report paired differences with Student-t intervals.

E. Problem Statement

A repair *strategy* π observes, at each step t, only the unlabeled incoming batch X_t and its own model state, and may rewrite any subset of blocks from the golden copy. Let $A_\pi(t)$ be the top-1 accuracy of its model on a held-out labeled set after step t, A_0 the clean accuracy, $D_\pi(T)$ the bytes it moved and $E_\pi(T)$ the energy it spent over a mission of T steps of length Δ. We evaluate

$$\bar{A}_\pi = \frac{1}{T}\sum_{t=1}^{T} A_\pi(t), \quad (2)$$

$$\mathcal{U}_\pi = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}[A_\pi(t) \geq A_0 - 0.05]\,, \quad (3)$$

$$\mathcal{L}_\pi = \Delta \sum_{t=1}^{T} \max(A_0 - A_\pi(t), 0)\,, \quad (4)$$

i.e. mean accuracy, availability (fraction of the mission within five points of clean) and accuracy-loss area in point-hours, alongside D_π and E_π, without labels or ground contact. No strategy need dominate on all criteria; periodic scrubbing with interval h traces a one-parameter family $\{(\bar{A}_h, D_h)\}$, the frontier any detection-based policy must beat.

V. THE AEGIS-V2 METHOD

Aegis-v2 runs once per monitoring period (one hour in our experiments) on the batch of images the satellite has just acquired. Fig. 6 gives the overview and Algorithm 1 the procedure.

A. Design Evolution: From Aegis-v1 to v2

The design was shaped by what did not work (Table IV). A first prototype (Aegis-v1) borrowed from plasticity research: it watched for "dead" low-firing neurons in a small multilayer perceptron, modeled faults as zeroed weights, and revived neurons with noise or re-initialization in the spirit of ReDo [35] and continual backpropagation [36]. A prior-art review showed three problems. The fault model was wrong: SEUs in float32 weights create huge values, not dead units, so a dead-neuron monitor watches for the opposite symptom. Re-initialization re-randomizes capacity but cannot recover a specific learned function without a training signal, and an inference-only payload has neither gradients nor labels. And a flat MLP does not represent the batch-normalized CNNs

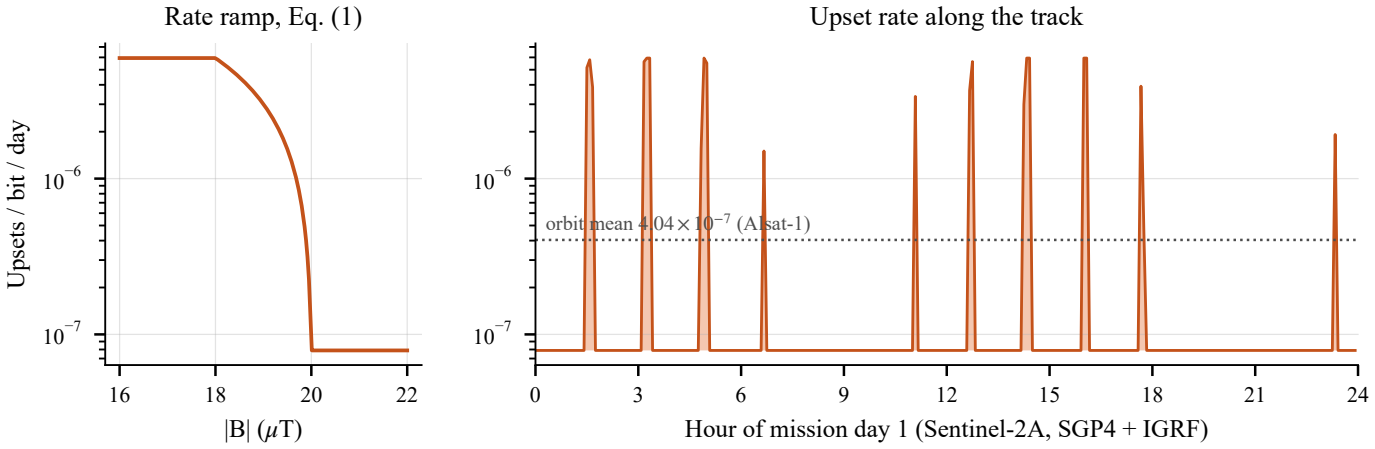


Fig. 4. Left: the rate ramp of Eq. (1). Right: the per-bit upset rate along the first mission day. Upsets arrive in short bursts during SAA passes over a low background; the dotted line is the orbit mean, calibrated to Alsat-1.

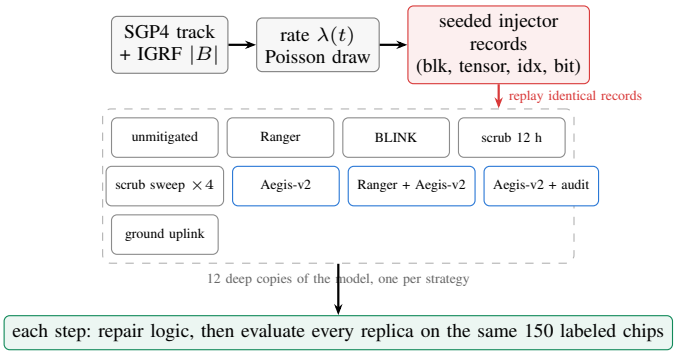


Fig. 5. Simulator data flow. One seeded injector samples each step's upsets; every replica replays the same records, so the strategies differ only in how they respond and all comparisons within a seed are paired.

TABLE IV
FROM AEGIS-V1 TO AEGIS-V2: WHAT CHANGED AND WHY.

Aspect	Aegis-v1	Aegis-v2 (inspiration)
Fault model	zeroed weights, dead neurons	uniform bit flips, Alsat-1 rates, real orbit [12]
Monitor	low-firing neuron counts	activation envelope (Ranger [14]) + BN drift (BLINK [16])
Confirmation	none	CRC-32 per block (RADAR-style [17])
Repair	noise / re-init (ReDo [35])	bit-exact restore from golden copy (scrubbing [8])
Model	MLP	ResNet-18, MobileNetV2 with BN
Evaluation	single-shot, synthetic inputs	3-day paired campaigns, real EuroSAT chips

flown in practice. Aegis-v2 keeps the idea of a self-monitoring network but replaces each piece with one grounded in the reliability literature.

B. Blocks

We partition θ into ordered *blocks*, the unit of detection, restoration and accounting: for ResNet-18 the stem, the eight residual blocks `layerX.Y` and the classifier (10 blocks, 38.7 kB to 18.9 MB); for MobileNetV2 the 19 `features.N` stages and the classifier (20 blocks). Table XV lists the ResNet-18 blocks with the faults each received and the restores each needed. A single function maps both parameter keys and module names to block identifiers; an early version used a separate mapping for monitor alarms whose fallback silently sent every MobileNetV2 alarm to the stem.

C. Tier 1: Activation Envelope

During calibration on $n \geq 100$ clean chips we record, for every convolution and linear layer l, the minimum m_l and maximum M_l of its output. The envelope is widened by a margin $\alpha = 0.5$ of the range:

$$[\ell_l, u_l] = [m_l - \alpha(M_l - m_l),\ M_l + \alpha(M_l - m_l)]. \tag{5}$$

At run time layer l raises an alarm if any output is NaN or $\pm\infty$ or falls outside $[\ell_l, u_l]$. Tier 1 catches exponent-MSB explosions immediately. The choice of envelope matters: an envelope of mean $\pm k\sigma$ fires on essentially every clean batch because activations are heavy-tailed, which silently degenerates the gate into "restore everything" (Section VIII).

D. Tier 2: Batch-Norm Drift

Each batch-norm layer stores running statistics (μ_c, σ_c^2) per channel c. For the input x of BN layer l on the current batch, with batch statistics $(\hat{\mu}_c, \hat{\sigma}_c)$, $s_c = \sqrt{\sigma_c^2 + \epsilon}$ and $\hat{s}_c = \sqrt{\hat{\sigma}_c^2 + \epsilon}$, the drift score is

$$d_l = \frac{1}{C} \sum_{c=1}^{C} \left(\frac{|\hat{\mu}_c - \mu_c|}{s_c} + \frac{|\hat{s}_c - s_c|}{s_c} \right), \tag{6}$$

set to a sentinel if any statistic is non-finite. The per-layer threshold is calibrated on clean batches of the run-time batch size as $\tau_l = \max(3.5,\ 2.5 \max_{\text{clean}} d_l)$. Tier 2 is sensitive to mid-exponent and large mantissa shifts that keep activations inside the T1 envelope.

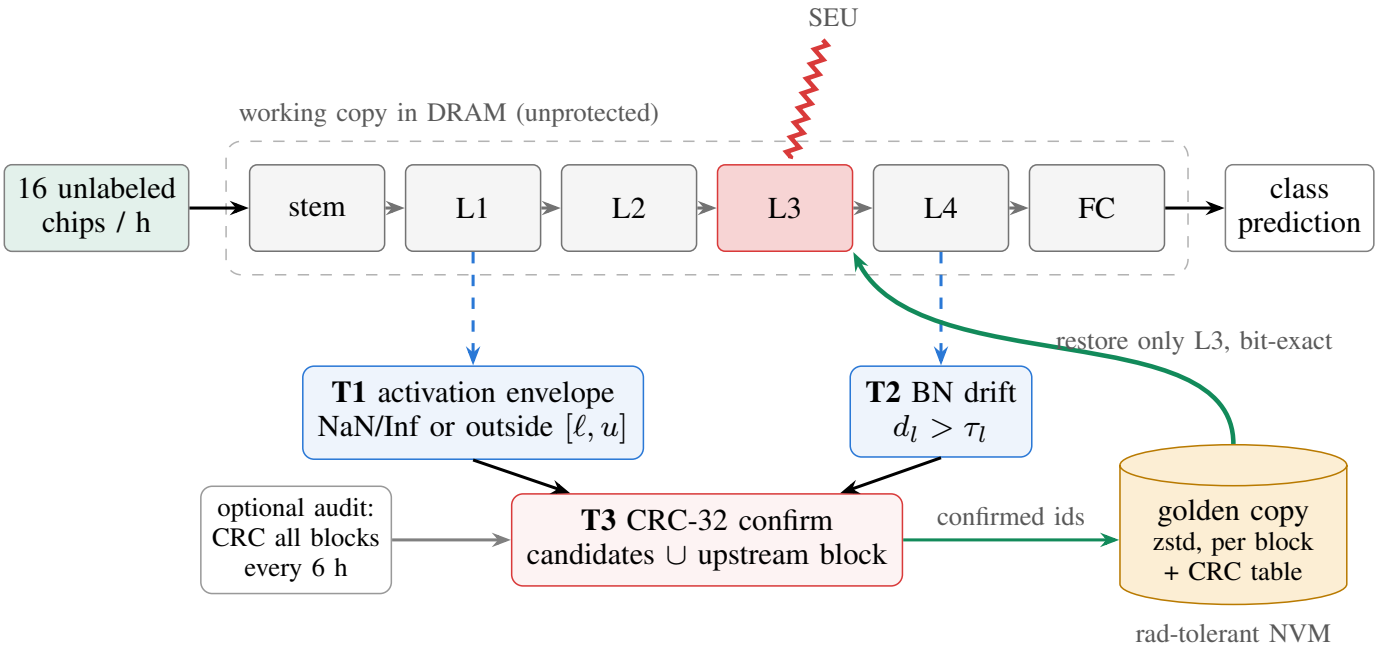


Fig. 6. Aegis-v2 repair loop. Hooks on each incoming batch feed the behavioral tiers T1 and T2, which propose candidate blocks without labels. T3 recomputes CRC-32 over each candidate and the block upstream of the earliest candidate, compares with golden checksums, and only blocks whose stored bits changed are restored from the compressed golden copy. Cascade victims downstream of a corrupted block raise alarms but pass T3 and are left untouched.

E. Tier 3: CRC Confirmation and the Upstream Rule

Behavioral alarms say where to *look*, not what is broken: an exponent flip in one block corrupts every activation downstream, so T1 and T2 typically flag a cascade of blocks. Tier 3 resolves this exactly. At calibration it stores the CRC-32 of each block's serialized parameters; at run time it recomputes the CRC of each candidate and declares a block corrupted iff the checksums differ. Cascade victims pass and are not restored.

T1 hooks convolution outputs and T2 hooks BN *inputs*, so a fault in a BN parameter at the end of block k, e.g. a sign flip on a running variance that produces NaN, first becomes visible at the hooks of block $k{+}1$. Aegis-v2 therefore always adds the block immediately upstream of the earliest candidate to the T3 set. Section VII-F measures what this rule is worth on identical fault streams.

F. Golden Store and Restore

The golden copy holds one zstd-compressed [43] safetensors blob per block. Trained float32 weights compress poorly ($44.77 \rightarrow 41.58$ MB for ResNet-18, $9.08 \rightarrow 8.44$ MB for MobileNetV2), so the store is essentially the model size (Table V). Restores are bit-exact, so no batch-norm recalibration [44] is needed; we keep it optional and off, and charge one inference per buffered image when enabled. An INT8 variant (per-output-channel symmetric quantization) shrinks the ResNet-18 store to 9.2 MB at a cost of 0.7 points on 1,000 chips; its restores are lossy, so restored blocks are re-baselined individually in T3. Refreshing all golden checksums while any block may be corrupt would bake the corruption into the reference and make it permanently invisible; Aegis-v2 only re-baselines blocks it has just restored.

Algorithm 1 Aegis-v2 step on an incoming batch

Input: blocks $b_1, \ldots, b_K$ in forward order; envelopes $[\ell_l, u_l]$; thresholds τ_l; golden CRCs g_k; golden store $\mathcal{G}$
Output: restored block set R

```
forward pass on the batch with T1/T2 hooks attached
C ← {block(l) : T1 alarm at l ∨ d_l > τ_l}
if C = ∅ then
    return ∅ {healthy: no checksum, no traffic}
end if
k* ← min{k : b_k ∈ C};  C ← C ∪ {b_{k*−1}} {upstream rule}
R ← {b_k ∈ C : CRC32(b_k) ≠ g_k}
for all b_k ∈ R in forward order do
    b_k ← decompress(G[k]) {bit-exact restore}
end for
return R
```

TABLE V
GOLDEN-STORE OPTIONS FOR RESNET-18.

Format	Size (MB)	Ratio	Restore
Raw float32	44.77	1.00×	bit-exact
zstd (level 3) per block	41.58	1.08×	bit-exact (default)
INT8 per-channel + scales	9.2	4.9×	lossy, −0.7 pts

G. CRC Audit for Sub-Threshold Faults

Behavioral detection cannot see a fault that keeps activations inside the envelope and statistics inside tolerance. In seed 1 a single weight changed from 0.106 to 1.7, cost 17 points of accuracy and raised no alarm. The *Aegis-v2 + CRC audit* variant therefore checksums every block on a fixed period (6 h

TABLE VI
SIMULATION CONSTANTS. M = MEASURED, C = CALIBRATED, A = ASSUMED.

Quantity	Value	
Orbit-mean upset rate (Alsat-1 [12])	4.04×10^{-7} /bit/day	M
Single-bit fraction	98.55%	M
Background / SAA-core rate	7.89×10^{-8} / 5.94×10^{-6}	C
SAA-core / background ratio	75.3	A
SAA ramp, $\lvert B\rvert$ edge $\rightarrow$ core	20,000 $\rightarrow$ 18,000 nT	A
SAA-core-equivalent time	79.9 min/day	C
Flash read / DRAM write / DRAM read	15 / 25 / 20 pJ/bit	A
Monitor elementwise op	6.9 pJ	A
Inference energy (Jetson-class proxy)	12.5 mJ/image	A
Uplink rate, contacts	1 Mbps, 4×480 s/day	A
RF receive energy	500 pJ/bit	A
T1 margin / T2 floor, factor / audit	0.5 / 3.5, 2.5 / 6 h	–

TABLE VII
PER-BATCH MONITORING COST (16 IMAGES, 224×224).

	ResNet-18	MobileNetV2
Inference MACs per image	1.82 G	≈0.30 G
T1/T2 hook ops per batch	278 M	748 M
Hook ops / inference MACs (batch)	1.0%	≈16%
Hook energy per batch	1.9 mJ	5.2 mJ
Full T3 scan (all blocks)	44.8 MB read	9.1 MB read
Full T3 scan energy	7.2 mJ	1.5 mJ

by default) and restores any mismatch, independently of the behavioral gate.

H. Combining with Range Restriction

Ranger-style clamping can run under Aegis-v2: monitor hooks are registered before the clamp hooks so that T1 observes raw activations, and the clamps limit damage in the interval before a restore. We evaluate this as *Ranger + Aegis-v2*.

I. Cost Model and Complexity

Every repairing strategy is charged for the bytes it moves and the energy that costs. A restore of block k with compressed size c_k and raw size r_k costs

$$E_{\text{restore}}(k) = 8\, c_k\, e_{\text{flash}} + 8\, r_k\, e_{\text{DRAM,w}}, \tag{7}$$

a T3 checksum of block k costs $8\, r_k\, e_{\text{DRAM,r}}$, and the T1/T2 hooks cost e_{op} per elementwise operation (four passes over each convolution/linear output, three over each BN input). $e_{\text{op}} = 6.9\,\text{pJ}$ comes from dividing a 12.5 mJ ResNet-18 inference on a Jetson-class device by its 1.82 G MACs; this treats an elementwise pass as one MAC and is likely pessimistic. Monitoring is charged only on the monitored incoming batch. The ground uplink receives at 1 Mbps during four 480 s contacts per day (≈ 60 MB per pass, as reported for Chaohu-1 [4]) at 500 pJ/bit. Ranger and BLINK move no bytes and their per-inference compute is not charged. Table VI lists every constant and Table VII the resulting per-batch costs.

The asymmetry in Table VII is worth noting: MobileNetV2's inverted residuals expand activations six-fold and add a batch-norm layer after every convolution, so elementwise monitoring is cheap relative to inference on ResNet-18 ($\approx 1\%$) but not on MobileNetV2 ($\approx 16\%$).

TABLE VIII
MODELS.

	ResNet-18	MobileNetV2
Stored values (incl. BN buffers)	11.19 M	2.27 M
Stored size (float32)	44.77 MB	9.08 MB
Blocks	10	20
Weights	public checkpoint	fine-tuned (ours)
Training split	not published	held out, verified
Clean accuracy (150 eval chips)	89.3%	99.3%
Clean accuracy (1,000-chip check)	92.8%	–

VI. EXPERIMENTAL SETUP

Data and splits. EuroSAT RGB chips are bicubically upsampled to 224×224 and normalized with EuroSAT channel statistics; logits follow the checkpoint's label order. Calibration (10 chips per class, 100 total) and labeled evaluation (15 per class, 150 total) sets are disjoint. Monitoring operates on a rotating window of 16 unlabeled chips per step; labels are used only to measure accuracy.

Models (Table VIII). ResNet-18 uses a public EuroSAT checkpoint (`cm93/resnet18-eurosat`) whose training split is not published, so its clean accuracy may be optimistic, though relative degradation is unaffected. To remove that caveat on a second architecture we fine-tuned an ImageNet-initialized [45] MobileNetV2 for three epochs (AdamW [46] with a one-cycle schedule [47], batch 64, learning rate 10^{-3}, flip and 90°-rotation augmentation) on 26,035 chips, holding out the 965 chips used by any campaign split; the simulator refuses any split not contained in the held-out set. Training took 159 s on a T4 GPU and reached 98.1% training accuracy.

Mission. Each campaign simulates three days in 72 one-hour steps for three radiation seeds, with the Sentinel-2A track and the rate model of Section IV. Per step the simulator injects the step's upsets, replays them on every replica, runs each strategy's repair logic, and evaluates every replica on the 150 labeled chips.

Strategies (Table IX). Ranger clamps activations to profiled bounds with a 0.1 margin [14]; BLINK renormalizes anomalous BN inputs to reference statistics (threshold 3.0) [16]; scrubbing restores the full model every 12 h, plus a sweep at 1, 3, 6 and 24 h; the ground uplink queues CRC-confirmed blocks and reloads them only during contacts, rate-limited by the link.

Metrics and statistics. We report mean accuracy $\bar{A}$, availability $\mathcal{U}$ and loss area $\mathcal{L}$ (Section IV), final accuracy (a single noisy snapshot), repair traffic in MB/day and modeled energy in mJ/day. Intervals are 95% Student-t over three seeds ($t_{0.975,2} = 4.30$), so they are wide; paired differences use the same fault stream per seed. The *bandwidth verdict* asks whether any scrubbing interval dominates a strategy (higher accuracy at no more traffic) and how much traffic scrubbing needs to match it, interpolating linearly between swept intervals.

TABLE IX
STRATEGIES COMPARED.

Strategy	Detects	Repairs weights	Store	Ground
Unmitigated	–	–	–	–
Ranger clipping	–	no (clamps)	–	–
BLINK re-centering	BN	no (re-centers)	–	–
Scrubbing (h)	–	all, every h	✓	–
Ground uplink	T1–T3	at next contact	–	✓
Aegis-v2	T1–T3	at next batch	✓	–
Aegis-v2 + audit	T1–T3, audit	at next batch	✓	–
Ranger + Aegis-v2	T1–T3	next batch, clamps	✓	–

Implementation. The simulator is written in PyTorch. The final campaigns ran on an NVIDIA T4 (about 200 s per seed for ResNet-18 with 12 replicas, 225 s for MobileNetV2). The repository includes 51 unit and regression tests, including regression tests for the pitfalls of Section VIII. Every table in Section VII is generated by a script from the result files.

VII. RESULTS

A. Accuracy Under Accumulating Upsets

Table X and Fig. 7 give the main results. On ResNet-18, with 147 upsets per day and 4.9 on the exponent MSB, the unprotected model is at chance for almost the entire mission (mean 14.0%, final 10.0% in every seed). Range clipping and BN re-centering slow the collapse but cannot reverse it, because the corrupted weights stay in memory and accumulate (means 38.5% and 34.0%, ending at 27.1% and 13.6%). Twelve-hour scrubbing returns the model to clean accuracy at each reload but spends most of each interval broken (mean 37.0%). The ground uplink repairs only during contacts (mean 66.2%). Aegis-v2 holds 87.4% against a clean 89.3%; with the CRC audit, 88.9%, with a final accuracy equal to clean in every seed.

Fig. 8 and Table XI break the result down by seed. In seeds 0 and 2 all three Aegis-v2 variants hold clean accuracy for the entire mission. The gap in the mean comes from a single event: at hour 49 of seed 1 a sub-threshold weight fault ($0.106 \rightarrow 1.7$) removed 17 points without raising any alarm and persisted to the end of the mission; the audit variant repaired it at the next six-hourly audit. The baselines vary far more between seeds (Ranger from 12.7% to 55.3%), because a single early exponent hit decides most of their mission.

Combining Ranger with Aegis-v2 gives identical accuracy here and 24% less traffic; its restore events are as frequent (63 versus 62 over all seeds), so the saving comes from restoring fewer or smaller blocks per event, and we have not isolated the mechanism. MobileNetV2 stores five times fewer bits and is hit five times less often (29 upsets/day). Its ReLU6 activations are bounded by construction, which plausibly explains why range clipping is far more effective on it (98.8%) than on ResNet-18. All three Aegis-v2 variants hold the clean 99.3% in every MobileNetV2 seed.

B. Paired Comparison

Because every strategy sees the same faults within a seed, Table XII reports paired differences. On ResNet-18 Aegis-v2 beats every non-Aegis-v2 baseline in all three seeds: by $+50.5$ points over 12-hour scrubbing, $+21.2$ over the ground uplink, $+48.9$ over range clipping, $+53.4$ over BN re-centering and $+73.4$ over no protection. On MobileNetV2 the differences are smaller and, except against BN re-centering, not significant with three seeds; in seed 2 no fault was damaging enough to separate any strategies.

C. Availability and Accuracy-Loss Area

Mean accuracy hides how the loss is distributed in time. Table XIII reports availability, the share of the mission spent within five points of clean accuracy, and the accuracy-loss area. On ResNet-18 Aegis-v2 is available 89% of the time and the audit variant 98%, against 26% for 12-hour scrubbing, 59% for the ground uplink and under 15% for clipping, re-centering and no protection. The loss areas differ by more than an order of magnitude: 139 point-hours for Aegis-v2 and 29 with the audit, against 1,663 for the uplink and 3,771 for scrubbing. On MobileNetV2 every Aegis-v2 variant is available for the whole mission with zero loss area, while scrubbing and the uplink lose 872 and 432 point-hours.

D. Accuracy per Byte: the Bandwidth Verdict

Scrubbing has one knob, the interval, which trades traffic for accuracy along the curve of Fig. 9. No interval dominates any Aegis-v2 variant on either architecture. Interpolating the frontier, scrubbing needs about 889 MB/day to reach Aegis-v2's 87.4% on ResNet-18, 4.7× the 188 MB/day Aegis-v2 moves; 975 MB/day to match the audit variant (3.5×); and the combination with Ranger reaches the same accuracy as Aegis-v2 at 6.2× less traffic than the matching scrub. On MobileNetV2 the ratio is about 60×, mostly because Aegis-v2 moves almost nothing (3.4 MB/day, nine restore events over all seeds) while the matching 1-hour scrub rewrites the model 24 times a day.

The comparison is at *equal accuracy*. In absolute terms Aegis-v2 on ResNet-18 moves more bytes than 12-hour scrubbing (Fig. 10): it restores on demand, about seven times a day, while the 12-hour scrub moves the model only twice; but the scrub buys 37% mean accuracy with those bytes and Aegis-v2 buys 87%.

E. Repair Activity and Where Faults Land

Fig. 11 shows the arrival process for one seed: upsets bunch into the consecutive daily SAA passes, and quiet hours alternate with bursts of 20–30 flips. Classifying every recorded hit by the relative change it caused (destructive: non-finite, exponent MSB, NaN-producing variance sign flip or more than $10^3\times$; moderate: more than $3\times$; otherwise negligible), only 3.4% of ResNet-18 hits are destructive and 77.9% negligible (Table III). Table XIV summarizes repair activity: Aegis-v2 restores in 29% of ResNet-18 steps, about seven events a day of 27 MB each, and in 4% of MobileNetV2 steps.

Table XV explains the traffic: faults land where the bits are, and on ResNet-18 the four `layer3`/`layer4` blocks hold 94% of the parameters, received 95% of the hits and account for

TABLE X
MISSION OUTCOMES OVER THREE SIMULATED DAYS (MEAN OF THREE SEEDS, 95% STUDENT-t INTERVALS). REPAIR TRAFFIC AND ENERGY ARE PER SIMULATED DAY; "N.C." = NOT CHARGED (STRATEGY MOVES NO BYTES).

	ResNet-18 (clean 89.3%, 147 flips/day)				**MobileNetV2** (clean 99.3%, 29 flips/day)			
Strategy	Mean acc. (%)	Final acc. (%)	MB/day	mJ/day	Mean acc. (%)	Final acc. (%)	MB/day	mJ/day
Aegis-v2	**87.4**±8.3	83.6±24.9	188.1	144.1±35.8	**99.3**±0.0	99.3±0.0	3.4	125.6±2.5
Aegis-v2 + CRC audit	**88.9**±1.7	89.3±0.0	279.6	202.3±20.3	**99.3**±0.0	99.3±0.0	18.1	136.4±2.5
Ranger + Aegis-v2	**87.4**±8.3	83.6±24.9	143.0	120.5±35.6	**99.3**±0.0	99.3±0.0	1.7	124.7±1.0
Periodic scrubbing (12 h)	37.0±27.0	89.3±0.0	83.2	27.9	87.2±26.1	99.3±0.0	16.9	5.7
Ground uplink reload	66.2±10.2	52.4±91.4	121.5	634.5±167.3	93.3±12.9	99.3±0.0	3.5	141.5±27.7
Ranger clipping	38.5±56.4	27.1±43.1	0	n.c.	98.8±1.4	96.7±6.0	0	n.c.
BLINK re-centering	34.0±42.0	13.6±16.8	0	n.c.	79.5±17.1	60.4±44.9	0	n.c.
Unmitigated	14.0±9.8	10.0±0.0	0	n.c.	76.9±51.7	39.8±128.1	0	n.c.
Scrubbing 1 h	89.3	–	998.0	334.6	99.3	–	202.4	67.9
Scrubbing 3 h	77.6	–	332.7	111.5	97.3	–	67.5	22.6
Scrubbing 6 h	66.4	–	166.3	55.8	96.4	–	33.7	11.3
Scrubbing 24 h	22.0	–	41.6	13.9	85.1	–	8.4	2.8

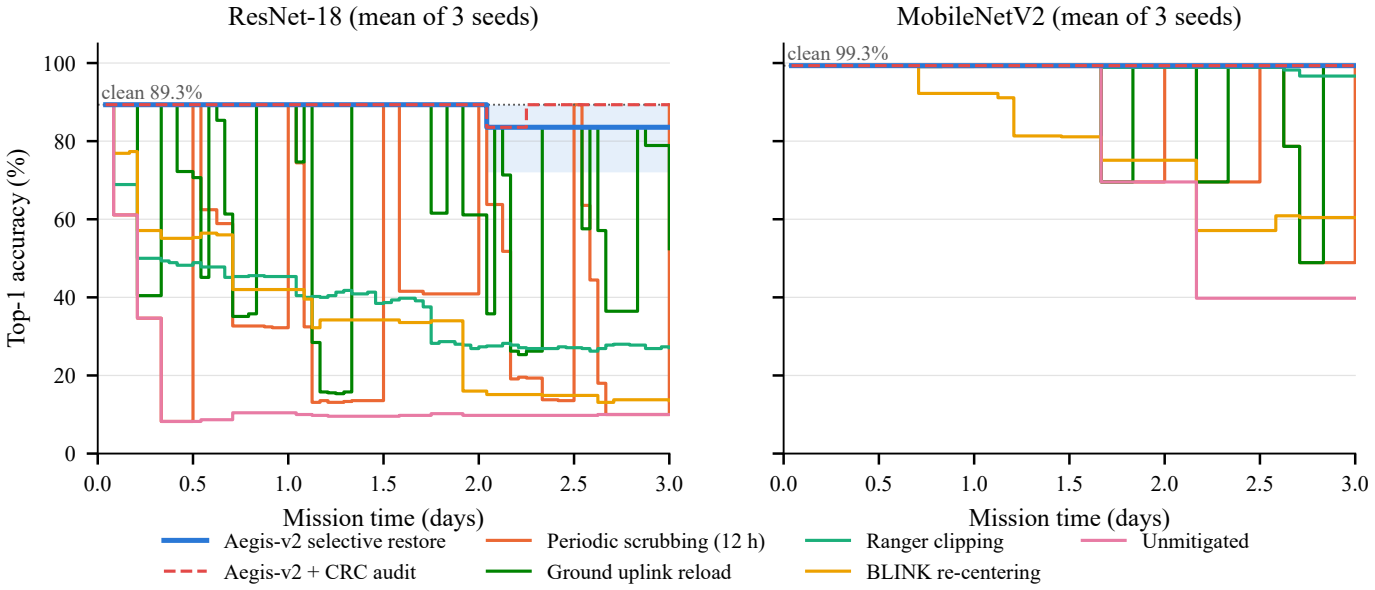


Fig. 7. Accuracy over the mission, averaged over three seeds (shaded: min–max across seeds for Aegis-v2). Unprotected ResNet-18 reaches chance (10%) within eight hours. Scrubbing recovers fully at each 12-hour reload and then decays as new upsets arrive; the ground uplink recovers only during contacts. Aegis-v2 stays at the clean line except where a sub-threshold fault in one seed escapes the behavioral gate (day 2); the CRC audit variant repairs it at the next audit.

92% of the 162 block restores. They also show that Aegis-v2 restores more often than destructive hits occur (`layer3.1`: 3 destructive hits, 40 restores): moderate faults that shift batch-norm statistics trigger T2 and are repaired too, and a block may be restored repeatedly as new hits arrive. Layer selectivity saves the most when damage is confined to small blocks; for a model whose parameters concentrate at the end, a restore is a large fraction of a full scrub, and the saving comes mainly from *not* restoring when nothing is wrong.

F. Ablation: the Upstream-Block Rule

Our first GPU campaign ran before the upstream rule existed, with the same seeds and the same injector. Its fault streams are identical to the final campaign's (same flip counts and exponent hits in every seed), and the baselines reproduce, so it isolates the rule exactly (Table XVI, Fig. 12). In seed 1, at hour 17 a sign flip on `layer3.0.bn2.running_var` produced NaN. Its symptoms first appeared at the hooks of the next block, whose checksum was intact, so nothing was restored and both Aegis-v2 replicas fell to 10%. Plain Aegis-v2 recovered an hour later, when an unrelated alarm happened to checksum the culprit; Ranger + Aegis-v2, whose clamps changed which layers alarmed, stayed at chance for 35 hours. With the rule, neither replica dropped. The rule raises Ranger + Aegis-v2 from 45.7% to 83.5% mean accuracy in that seed and costs one extra checksum per alarm.

G. Single-Fault Detection Study

The campaigns show what Aegis-v2 achieves; they do not show what each tier sees. We therefore injected one persistent upset at a time into the clean model, 30 trials per bit field and architecture (300 in total), placed like the campaign injector

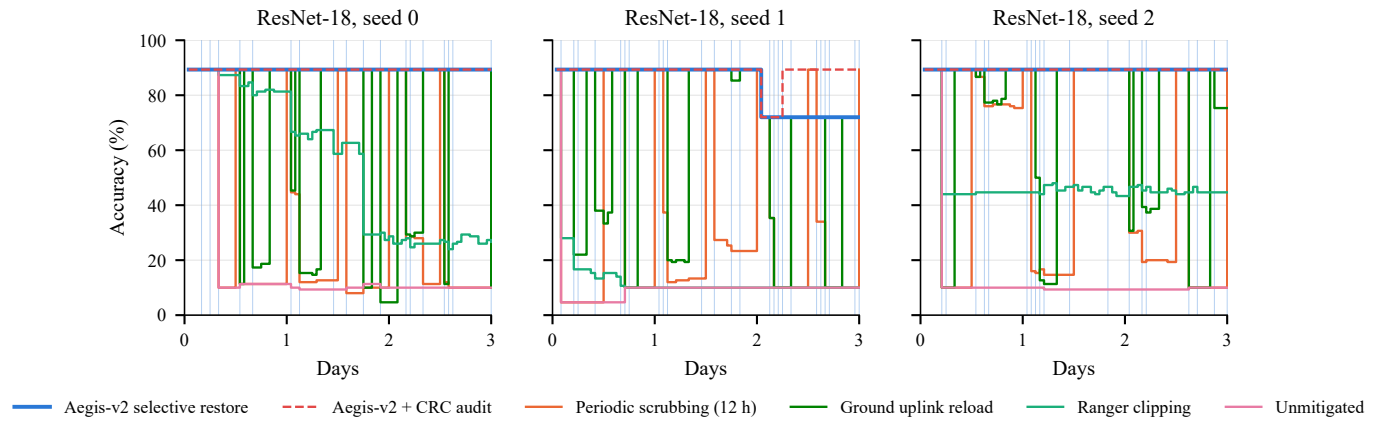


Fig. 8. Per-seed accuracy for ResNet-18 (faint vertical lines: steps in which Aegis-v2 restored at least one block). In seed 1 an undetected sub-threshold fault at hour 49 costs plain Aegis-v2 17 points until the end; the audit variant recovers within six hours.

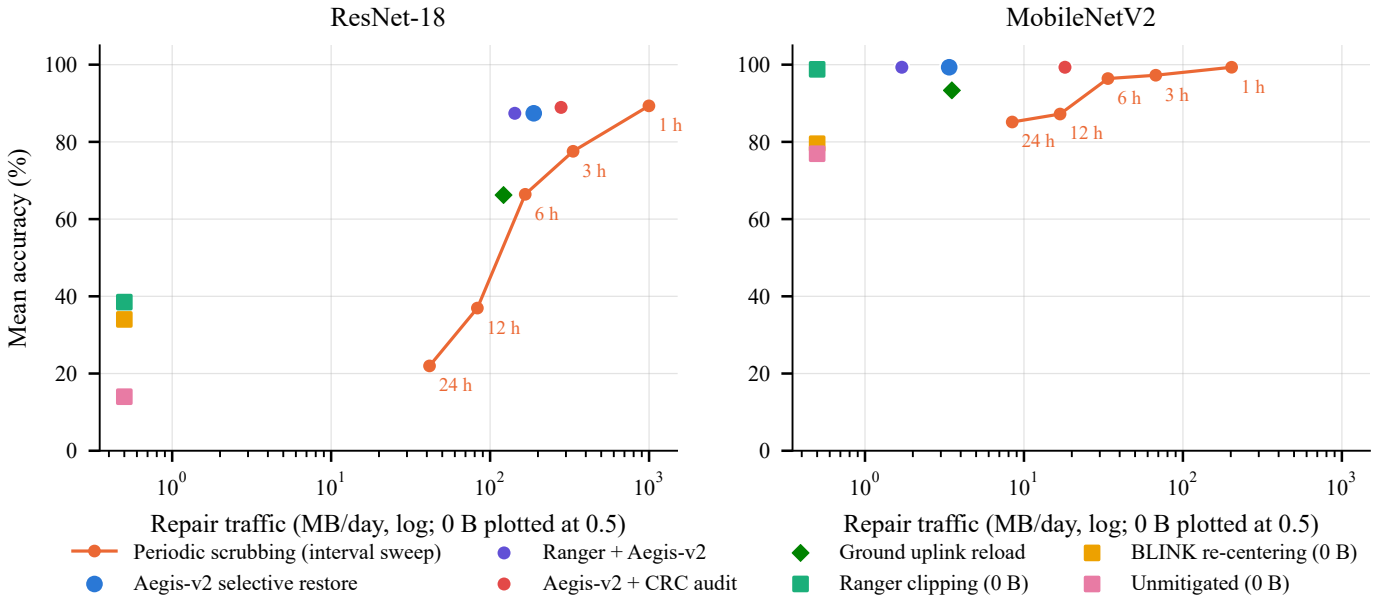


Fig. 9. Mean accuracy versus repair traffic. The orange curve sweeps the scrubbing interval from 24 h to 1 h; strategies that move no bytes are drawn at 0.5 MB/day. All Aegis-v2 variants lie above and to the left of the scrubbing frontier on both architectures.

(block and tensor in proportion to their bits) but with the bit field controlled. Each trial runs one monitored 16-image batch, records which tiers fired and which blocks T3 confirmed, measures the accuracy loss on the 150 labeled chips, and resets the model (Table XVII, Fig. 13).

Three findings stand out. *Damage is concentrated in one bit.* Only exponent-MSB flips caused any loss: 12 of 30 on each architecture, with losses up to 81 points on ResNet-18 and 89 on MobileNetV2. None of the 240 flips in the other fields lost a single chip, consistent with Table I. Weighting by field size, about 1.3% of uniformly placed flips are damaging ($\frac{1}{32} \times 40\%$), the same order as the 3% failure rate reported for FP32 flips in vision transformers [11]. *Every damaging flip was detected and localized.* Recall was 12/12 on both models, T3 confirmed the struck block in every alarmed exponent-MSB trial, and no trial restored a block that had not been struck. The 18 non-damaging exponent-MSB flips per model also raised alarms and were restored; they are real corruptions that happened not to cost accuracy on these chips, and restoring them prevents latent damage. *Behavioral tiers are noisy, the checksum is not.* On ResNet-18, 12.5% of the 120 harmless trials raised a T1 alarm, always at `layer4.0`, independent of where the fault was: some clean batches exceed the calibrated envelope of that layer. T3 checked `layer4.0` and its upstream block, found both intact, and restored nothing (in the two sign-flip trials where the struck block happened to be one of those two, it was correctly restored). MobileNetV2 raised no such alarms. The CRC gate is therefore what keeps an imperfect behavioral monitor cheap: its false alarms cost a checksum of two blocks (≈ 3 mJ on ResNet-18), never a restore, which also explains the large T3 share of ResNet-18's energy (Table XIX).

The study does not contradict the sub-threshold miss in the campaign (Fig. 8). That fault ($0.106 \rightarrow 1.7$, a factor of 16) came from an exponent bit below the MSB and kept

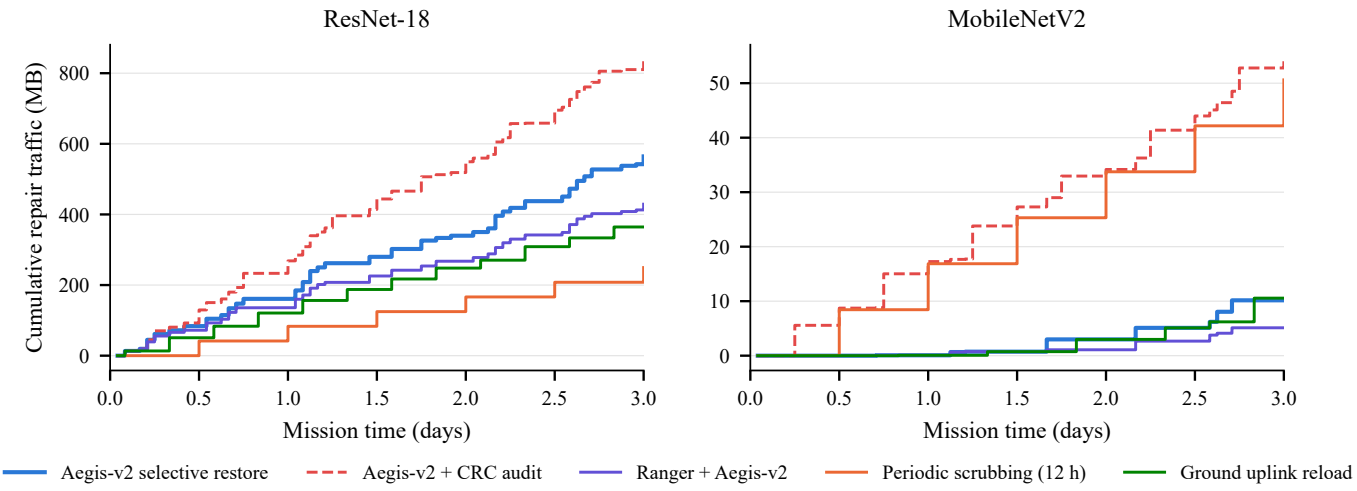


Fig. 10. Cumulative repair traffic (mean of seeds). On ResNet-18 Aegis-v2 restores on demand and moves more bytes than 12-hour scrubbing, at far higher accuracy (Table X); on MobileNetV2 it moves less than any timer-based scrub.

TABLE XI
PER-SEED MEAN ACCURACY IN %, WITH FINAL ACCURACY IN PARENTHESES.

Strategy	Seed 0	Seed 1	Seed 2
ResNet-18			
Aegis-v2	89.3 (89)	83.5 (72)	89.3 (89)
Aegis-v2 + audit	89.3 (89)	88.1 (89)	89.3 (89)
Ranger + Aegis-v2	89.3 (89)	83.5 (72)	89.3 (89)
Scrubbing (12 h)	31.2 (89)	30.2 (89)	49.5 (89)
Ground uplink	66.1 (10)	62.2 (72)	70.4 (75)
Ranger clipping	55.3 (27)	12.7 (10)	47.5 (45)
BLINK re-centering	52.7 (9)	19.8 (10)	29.6 (21)
Unmitigated	17.9 (10)	10.0 (10)	14.1 (10)
MobileNetV2			
Aegis-v2	99.3 (99)	99.3 (99)	99.3 (99)
Aegis-v2 + audit	99.3 (99)	99.3 (99)	99.3 (99)
Ranger + Aegis-v2	99.3 (99)	99.3 (99)	99.3 (99)
Scrubbing (12 h)	80.7 (99)	81.6 (99)	99.3 (99)
Ground uplink	90.6 (99)	90.0 (99)	99.3 (99)
Ranger clipping	98.8 (95)	98.2 (96)	99.3 (99)
BLINK re-centering	83.9 (50)	71.6 (50)	83.1 (81)
Unmitigated	73.3 (10)	58.2 (10)	99.3 (99)

TABLE XII
PAIRED DIFFERENCE IN MEAN ACCURACY, AEGIS-V2 MINUS BASELINE (POINTS, 95% INTERVAL), AND SEEDS WHERE AEGIS-V2 WAS HIGHER.

	ResNet-18		MobileNetV2	
Baseline	Δ	wins	Δ	wins
Scrubbing (12 h)	$+50.5 \pm 23.6$	3/3	$+12.1 \pm 26.1$	2/3
Ground uplink	$+21.2 \pm 5.3$	3/3	$+6.0 \pm 12.9$	2/3
Ranger clipping	$+48.9 \pm 48.2$	3/3	$+0.5 \pm 1.4$	2/3
BLINK re-centering	$+53.4 \pm 36.4$	3/3	$+19.8 \pm 17.1$	3/3
Unmitigated	$+73.4 \pm 4.8$	3/3	$+22.4 \pm 51.7$	2/3
Aegis-v2 + audit	-1.5 ± 6.6	0/3	0.0 ± 0.0	0/3

every activation inside its envelope. None of our 30 sub-MSB exponent trials reproduced such a loss on a clean model: a sixteen-fold change to a weight that matters is rare under uniform placement, and in the campaign it arrived on a model that had already absorbed dozens of upsets. Rare, behaviorally silent faults of this kind are exactly what the periodic audit is for.

TABLE XIII
AVAILABILITY $\mathcal{U}$ (% OF MISSION WITHIN 5 POINTS OF CLEAN, 95% INTERVAL) AND ACCURACY-LOSS AREA $\mathcal{L}$ (POINT-HOURS, MEAN OF SEEDS).

	ResNet-18		MobileNetV2	
Strategy	$\mathcal{U}$ (%)	$\mathcal{L}$	$\mathcal{U}$ (%)	$\mathcal{L}$
Aegis-v2	89±48	139	100±0	0
Aegis-v2 + audit	98±10	29	100±0	0
Ranger + Aegis-v2	89±48	139	100±0	0
Scrubbing (12 h)	26±22	3771	85±32	872
Ground uplink	59±29	1663	93±16	432
Ranger clipping	8±22	3660	100±0	39
BLINK re-centering	14±47	3985	44±61	1426
Unmitigated	6±10	5424	75±58	1613

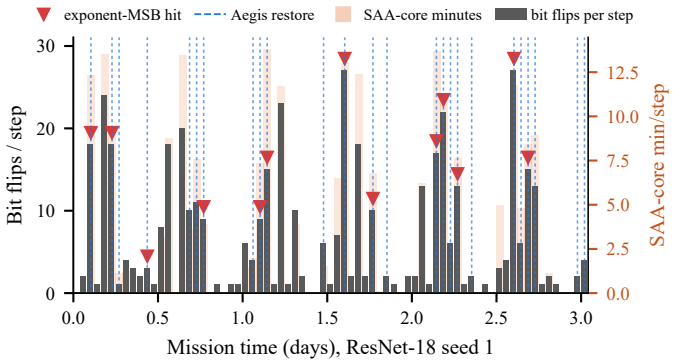


Fig. 11. Upsets per one-hour step for ResNet-18 seed 1 against SAA-core minutes in the step. Upsets cluster in SAA passes; exponent-MSB hits (triangles) arrive about five times a day; dashed lines mark steps where Aegis-v2 restored at least one block.

H. What a Damaged Classifier Predicts

For an Earth-observation product the failure mode matters as much as the failure rate. Fig. 14 pools the confusion matrices of the eight ResNet-18 trials that lost more than 20 points. Damage does not produce structured semantic confusions such

TABLE XIV
REPAIR ACTIVITY (MEAN PER SEED OVER THREE DAYS).

	ResNet-18	MobileNetV2
Restore events / seed	20.7	3.0
Block restores / seed	54.0	9.3
Steps with a restore (%)	29	4
Mean MB per restore event	27.3	3.4
Ranger + Aegis-v2 restore events / seed	21.0	3.0
Audits / seed (6 h)	12	12
Full scrubs / seed (12 h)	6	6
Uplink contacts used / seed	10.7	3.0
Golden store (MB, zstd)	41.58	8.44
Restore events per day	6.9	1.0

TABLE XV
RESNET-18 BLOCKS: SIZE, SHARE OF STORED BITS, RECORDED HITS, DESTRUCTIVE HITS AND AEGIS-V2 RESTORES (THREE SEEDS).

Block	MB	Bits (%)	Hits	Destr.	Restores
`stem`	0.04	0.1	0	0	0
`layer1.0`	0.30	0.7	9	1	1
`layer1.1`	0.30	0.7	11	0	1
`layer2.0`	0.92	2.1	17	1	3
`layer2.1`	1.18	2.6	28	2	7
`layer3.0`	3.68	8.2	110	7	13
`layer3.1`	4.73	10.6	148	3	40
`layer4.0`	14.70	32.8	417	17	57
`layer4.1`	18.89	42.2	583	14	39
`fc`	0.02	0.0	1	0	1

as River↔Highway; it collapses the classifier toward one class that depends on the fault. In those trials a single class absorbed 56–82% of all predictions (Highway in three trials, River in two, Annual Crop, Herbaceous Vegetation and Pasture in one each); two MobileNetV2 trials collapsed completely, labelling all 150 chips as Sea/Lake and as Highway respectively. A downstream consumer would see a confident, plausible-looking stream of one land-cover class, which is why silent corruption must be detected onboard rather than inferred from the product.

I. Transient Faults Do Not Trigger Restores

A behavioral monitor alone cannot distinguish a transient glitch in an activation buffer from corrupted weights. We injected exponent-MSB flips into activations, not weights, in 40 single-batch trials on ResNet-18 (Table XVIII). Tier 1 or 2 raised an alarm in 27.5% of trials; Tier 3 found no checksum mismatch in any, so no bytes were moved. This is the second purpose of the CRC gate: beyond localizing cascades, it makes repair decisions robust to faults that need no repair.

J. Energy: a Negative Result

Fig. 15 separates the two costs. Under our energy model Aegis-v2 spends 144 mJ/day on ResNet-18 and 126 mJ/day on MobileNetV2, versus 27.9 and 5.7 mJ/day for 12-hour scrubbing: 5.2× and 22× more. Fig. 16 and Table XIX show why. On ResNet-18, restores are 44% of the energy, T3 checksum reads 24% and the T1/T2 hooks 32%; on MobileNetV2 the hooks are 99%, because restores are rare and its expanded activations make elementwise monitoring

TABLE XVI
ABLATION OF THE UPSTREAM-BLOCK RULE ON IDENTICAL FAULT STREAMS (RESNET-18). MEAN ACCURACY (%) WITHOUT / WITH THE RULE; LAST TWO COLUMNS: FINAL ACCURACY OF RANGER + AEGIS-V2.

		Aegis-v2		+ Ranger		final (+R)	
Seed	Flips	w/o	with	w/o	with	w/o	with
0	442	89.3	89.3	89.3	89.3	89.3	89.3
1	461	82.5	83.5	45.7	83.5	72.0	72.0
2	421	89.3	89.3	89.3	89.3	89.3	89.3
mean	–	87.0	87.4	74.8	87.4	–	–

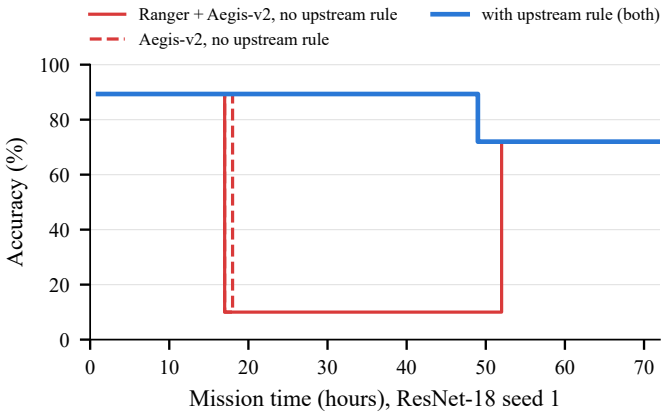


Fig. 12. Seed 1 of ResNet-18 with and without the upstream rule, same fault stream. A BN-variance sign flip at hour 17 sends both replicas to chance without the rule; Ranger + Aegis-v2 stays there for 35 hours.

expensive (Table VII). Our first prototype claimed a 98.9% energy saving because it did not charge the monitor; that claim was wrong. With the monitor charged, Aegis-v2 is an energy *cost*, and our claim is about bandwidth and availability.

The result needs perspective. The satellite spends energy on inference anyway: 16 images per hour at the 12.5 mJ ResNet-18 proxy is 4.8 J/day, so Aegis-v2's 144 mJ/day is 3% of the inference budget and 12-hour scrubbing's 28 mJ/day is 0.6%. Two further caveats cut in Aegis-v2's favor: 6.9 pJ per elementwise operation is likely pessimistic for fused hardware reductions, and monitoring could run on every n-th batch, trading detection latency for energy linearly. The ground uplink is the most expensive option by far (635 mJ/day on ResNet-18) because radio reception costs more than an order of magnitude more per bit than memory.

K. Cross-Architecture Observations

The two architectures agree on the ordering of the Aegis-v2 variants and on the failure of blind timers, and disagree on the baselines. MobileNetV2 is hit five times less often, so its unprotected mean (76.9%) is far above ResNet-18's, and range clipping nearly matches Aegis-v2 on it. The monitoring cost moves the other way: elementwise hooks are 16% of MobileNetV2's inference arithmetic against 1% for ResNet-18. A compact model with bounded activations therefore shifts the balance toward cheap in-place mitigation, while a large model with unbounded activations makes detection-gated repair indispensable.

TABLE XVII
SINGLE-FAULT DETECTION STUDY (30 TRIALS PER BIT FIELD AND ARCHITECTURE). DAMAGING = LOSES MORE THAN ONE EVALUATION CHIP. RECALL = DAMAGING TRIALS WITH A T1 OR T2 ALARM. LOCALIZED = ALARMED TRIALS IN WHICH T3 CONFIRMED THE STRUCK BLOCK. FALSE RESTORES = BLOCKS RESTORED THAT WERE NOT STRUCK.

Bit field	Damaging	Loss (pts)	T1 (%)	T2 (%)	Recall	Localized (%)	False restores
ResNet-18 (clean 89.3%, 150 trials)							
Exponent MSB (30)	12/30	18.2	100	100	12/12	100	0
Exponent (23–29)	0/30	0.0	13	0	–	0	0
Sign (31)	0/30	0.0	13	0	–	50	0
Mantissa high (16–22)	0/30	0.0	10	0	–	0	0
Mantissa low (0–15)	0/30	0.0	13	0	–	0	0
MobileNetV2 (clean 99.3%, 150 trials)							
Exponent MSB (30)	12/30	14.0	100	97	12/12	100	0
Exponent (23–29)	0/30	0.0	0	0	–	–	0
Sign (31)	0/30	0.0	0	0	–	–	0
Mantissa high (16–22)	0/30	0.0	0	0	–	–	0
Mantissa low (0–15)	0/30	0.0	0	0	–	–	0

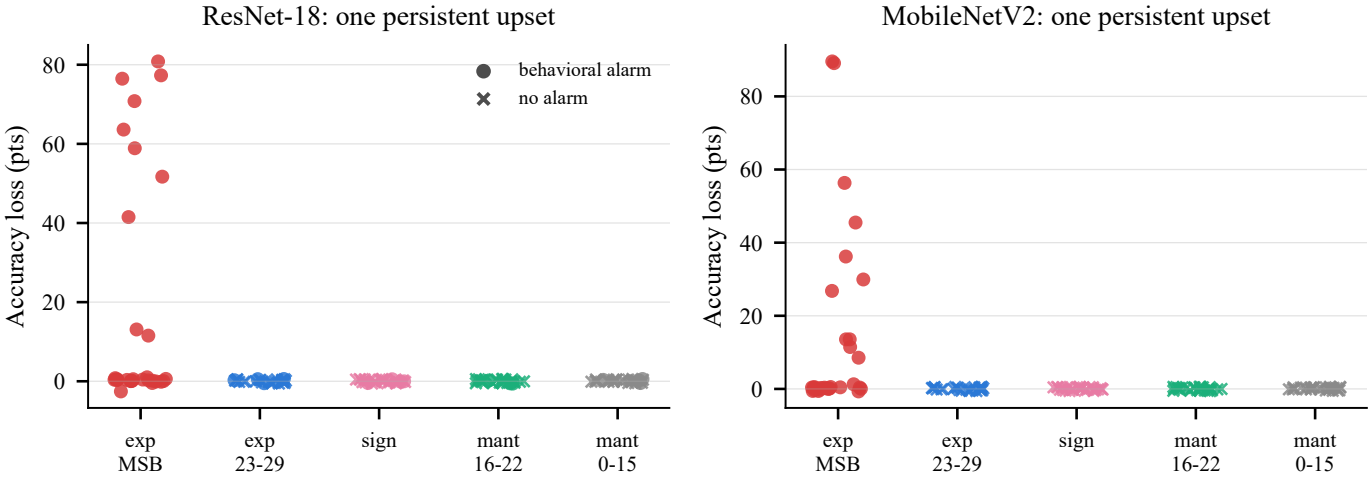


Fig. 13. Accuracy loss of single persistent upsets by bit field (jittered). Circles: a behavioral tier raised an alarm; crosses: no alarm. Only exponent-MSB flips cause damage, and every damaging flip was detected on both architectures.

TABLE XVIII
TRANSIENT ACTIVATION FAULTS (RESNET-18).

Trials (bit 30 flipped in one activation)	40
Behavioral alarm (T1 or T2)	27.5%
Spurious restores (T3 confirmed a block)	0.0%
Batch accuracy: clean / faulty / faulty + Ranger	89.7 / 88.6 / 89.7%

TABLE XIX
AEGIS-V2 ENERGY BY COMPONENT (MJ/DAY, SHARE OF TOTAL).

	Restore	T3 CRC	T1/T2 hooks	Total
ResNet-18	63.1 (44%)	34.9 (24%)	46.1 (32%)	144.1
MobileNetV2	1.1 (1%)	0.6 (0%)	123.9 (99%)	125.6

VIII. LESSONS: PITFALLS THAT INVALIDATED RESULTS

Table XX lists the mistakes that each silently produced a plausible but wrong result during this work, with the invariant that now prevents them. We list them because each is easy to repeat elsewhere; the fixes are covered by regression tests in the repository.

IX. DISCUSSION

Why gating works. Most upsets are harmless: 72.5% hit mantissas and 77.9% change a weight by less than a factor of three. A blind scrubber pays for all of them; an in-place mitigator pays for none of them but cannot undo the rare catastrophic one. Aegis-v2 pays only when behavior *and* checksums agree that something is broken, which is what lets it sit above the scrubbing frontier.

When it does not. The behavioral gate has a blind spot, faults large enough to matter but too small to move activations past calibrated bounds (Section VII-G). The audit closes it at a cost that scales with model size and audit frequency; a mission could schedule audits after SAA exits, when new faults are most likely, instead of on a fixed clock. Gating also cannot help when monitoring itself dominates energy (MobileNetV2), and it cannot repair the golden copy, whose integrity rests on rad-tolerant storage and its own ECC.

Choosing a strategy. Table XXI distills the results into guidance. If uplink and memory traffic are the scarce resources and availability matters, detection-gated repair with an audit is the best of the evaluated options. If energy is the binding constraint and long downtime is tolerable, a slow scrub is

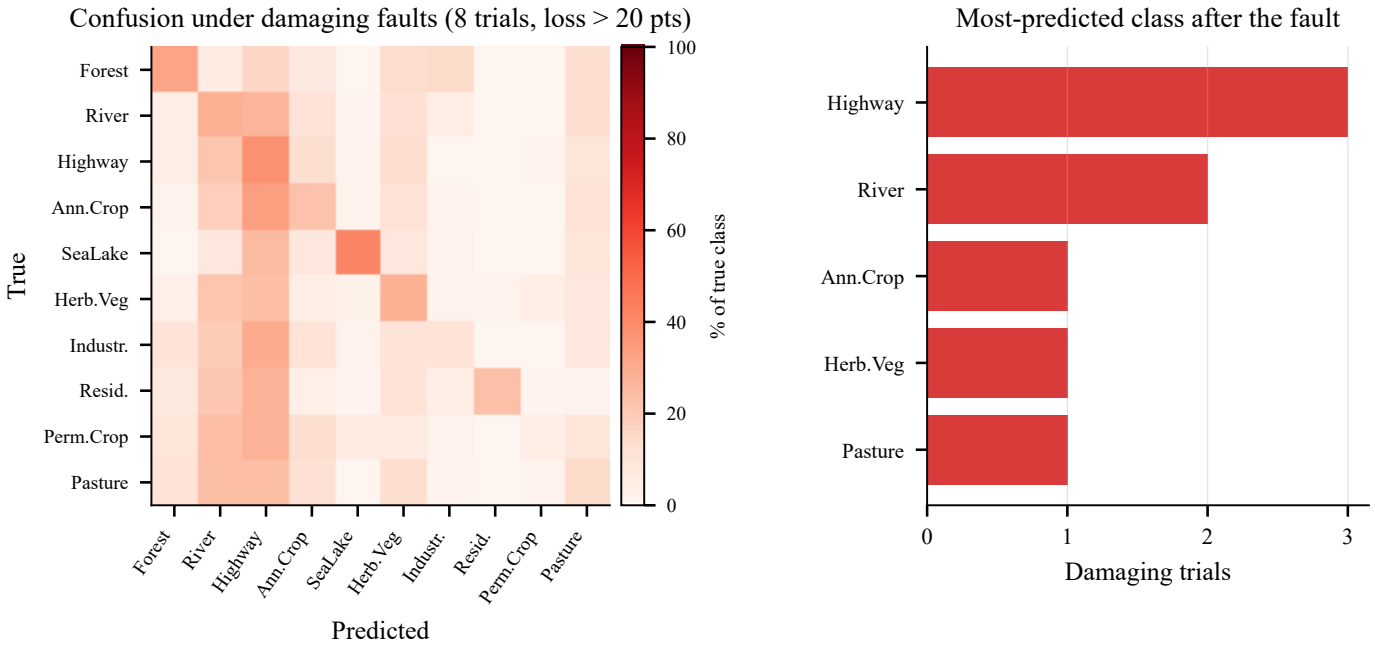


Fig. 14. Left: pooled row-normalized confusion matrix of ResNet-18 under the eight single upsets that cost more than 20 points. Right: the class that received the most predictions in each of those trials. A damaging upset collapses the classifier toward a fault-dependent class rather than producing semantically structured confusions.

TABLE XX
EVALUATION PITFALLS ENCOUNTERED DURING THIS WORK.

Pitfall	Symptom	Cause	Invariant
Synthetic inputs	"98.9% energy saving" headline	evaluation on Gaussian-noise tensors, always-on gate, uncharged monitor	never evaluate on random tensors; real EuroSAT chips only
Input contract	$\approx 18\%$ clean accuracy; chance with wrong labels	64 px input; alphabetical label order	224 px bicubic, dataset statistics, checkpoint label order
Envelope statistics	restores on every step	mean$\pm k\sigma$ bounds on heavy-tailed activations	observed extrema + margin over ≥ 100 chips
Calibration leakage	optimistic detection	calibration and evaluation chips overlap	disjoint splits; trained models exclude both
Poisoned reference	fault becomes undetectable	golden CRCs refreshed while corrupt	re-baseline only blocks just restored
Hook placement	35 h at chance (Table XVI)	BN fault seen one block downstream	always checksum the upstream block
Name mapping	every MobileNetV2 alarm sent to stem	two diverging layer-to-block mappings	one mapping function for keys and modules
Fault field	inflated severity	exponent-only injection ($4\times$ MSB rate)	uniform over all 32 bits; disclose any restriction
Uncharged costs	energy cost reported as saving	monitor, CRC or recalibration not charged	charge hooks, CRC reads, recalibration inferences

cheaper. If the model is small with bounded activations and no golden store is available, range clipping already recovers most of the accuracy.

Deployment. The golden copy fits comfortably in MRAM or rad-hard NOR for models of this size, or in 9.2 MB with INT8 at a small accuracy cost. T3 needs only a CRC engine and a table of 32-bit references; T1/T2 need reductions over activations that many accelerators can fuse into the layers that produce them. The method is complementary to ECC: with ECC on the working memory, Aegis-v2 would act on the residue of uncorrectable and logic-induced errors that beam studies show ECC misses [20].

TABLE XXI
WHICH STRATEGY FOR WHICH CONSTRAINT (FROM THE EVALUATED CAMPAIGNS).

Binding constraint	Best evaluated option
Availability, bandwidth	Aegis-v2 + CRC audit (98% / 100% available)
Bandwidth, low energy	Ranger + Aegis-v2 (lowest traffic at top accuracy)
Energy only	slow scrub (12–24 h); accepts long downtime
No golden store, compact model	range clipping (98.8% on MobileNetV2)
No onboard repair possible	ground uplink (only during contacts)

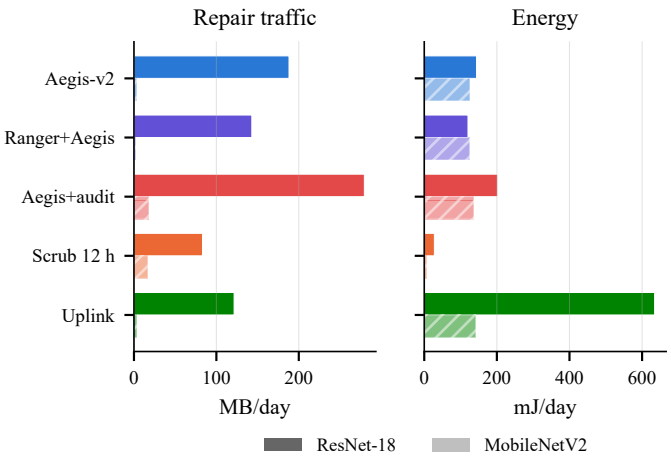


Fig. 15. Repair traffic and modeled energy per day. Solid: ResNet-18; hatched: MobileNetV2.

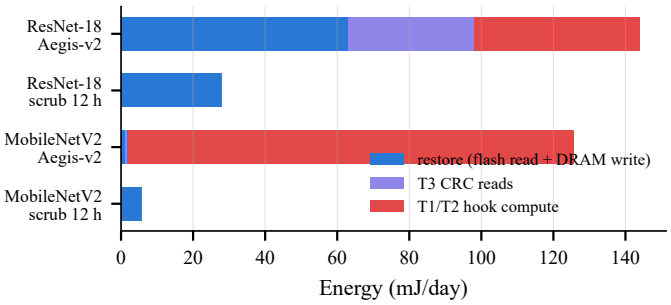


Fig. 16. Where Aegis-v2's energy goes, against 12-hour scrubbing.

X. Threats to Validity

Radiation model. Rates are transferred from Alsat-1 (686 km, 2002-era SRAM) to Sentinel-2A at 786 km and to modern memory, whose sensitivity can differ by orders of magnitude. The SAA-to-background ratio is assumed, the IGRF ramp is a proxy for a trapped-proton flux model such as AP8/AE9 combined with device cross-sections, solar-cycle variation is not modeled, and faults are injected only into parameters, not into control logic or activation buffers, which beam studies show can dominate [20]. **Statistics.** Three seeds and 150 evaluation chips give wide intervals; one chip is 0.67 points. The MobileNetV2 paired differences are mostly not significant. **Energy.** All energy constants are analytic proxies, not measurements on flight hardware, and Ranger/BLINK compute is not charged, which flatters those baselines. **Models.** The ResNet-18 checkpoint's training split is unknown; MobileNetV2 has a verified held-out split. **Mission length.** Three days captures several SAA cycles and dozens of exponent-MSB hits, but not long-term effects such as total ionizing dose. **Ablation.** The pre-fix campaign predates other small code changes; its fault streams and baselines match the final campaign, but its numbers outside the ablated strategies should not be cited. **Literature.** Several closely related works are 2026 preprints; we checked their records but cite their results as reported.

XI. Future Work

Natural extensions are (i) orbit-aware scheduling of audits and monitoring duty cycle around SAA passes, which targets both the blind spot and the energy cost; (ii) a physics-based environment (AP8/AE9 with device cross-sections via SPENVIS or CREME96); (iii) measured energy and latency on a Jetson- or NPU-class board; (iv) activation and logic faults alongside weight faults, with neutron-beam validation; (v) vision transformers and larger EO tasks such as segmentation; and (vi) longer missions spanning a solar-cycle phase.

XII. Conclusion

Onboard classifiers in LEO accumulate silent, sometimes catastrophic weight corruption. We showed that combining cheap behavioral signals with exact checksums lets a satellite repair only what is broken, when it is broken: on an orbit-calibrated, paired simulation of a Sentinel-2A mission, Aegis-v2 kept a ResNet-18 within two points of clean accuracy and available 89% of the time (98% with the audit), where 12-hour scrubbing, range clipping, re-centering and ground reload lost 23 to 55 points, and it did so with several times less repair traffic than the scrubbing interval that would match it. An exact ablation showed that one small rule, checksumming the block upstream of an alarm, prevents day-long outages. The same analysis showed what the method does not do: with honest accounting, always-on monitoring costs more energy than a slow scrub. We hope the simulator, the pitfalls and the results help the next design start from realistic faults and complete budgets.

Acknowledgment

The author thanks the maintainers of EuroSAT, PyTorch, sgp4, ppigrf and CelesTrak, whose open data and software made this study possible.

Appendix A
Reproducibility

All results come from `results/main_final`, `results/main_final_mobilenet`, `results/main_gpu` (ablation), `results/transient_faults.json` and `results/detection_study_*.json`. To regenerate:

```
python experiments/run_campaign.py --days 3 \
  --seeds 0 1 2 --eval-per-class 15 \
  --scrub-sweep-hours 1 3 6 24 --name main_final
python experiments/train_mobilenet.py --epochs 3
python experiments/run_campaign.py ... \
  --arch mobilenet_v2 --name main_final_mobilenet
python experiments/transient_faults.py --trials 40
python experiments/detection_study.py --arch resnet18
python experiments/detection_study.py --arch mobilenet_v2
python researchpaper/make_paper_figures.py
python researchpaper/make_paper_tables.py
```

Every seed file records the fault field, orbit model, golden-store format and recalibration flag with its numbers.

APPENDIX B
NOTATION

θ, N, B	stored model state, values, bits
b_k, K	block k in forward order, number of blocks
$\lambda(t)$, λ_{bg}, λ_{SAA}	upset rate; background; SAA core
B_{e}, B_{c}	field strength at SAA edge and core
$[\ell_l, u_l]$, α	T1 envelope of layer l, margin
d_l, τ_l	T2 drift score and threshold of BN layer l
g_k, $\mathcal{G}$	golden CRC of block k, golden store
$A_\pi(t)$, A_0	accuracy of π at step t; clean accuracy
$\bar{A}_\pi$, $\mathcal{U}_\pi$, $\mathcal{L}_\pi$	mean accuracy, availability, loss area
D_π, E_π	bytes moved, energy spent

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